

RJ45 & Transformer

The diagram illustrates the electrical connection between an RJ45 connector (J1) and a transformer (T1, PULSE_H1197). The RJ45 connector is connected to the transformer's primary windings (1-2, 3-4) and secondary windings (5-6, 7-8). The transformer is powered by +3.3V and GND through resistors R14, R16, R17, and R26. The output is connected to a BEAD (L2) and a 0.1uF capacitor (C18) to GND.

Component List:

Ref	Value	Part
J1	RJ45-LED	Connector
T1	PULSE_H1197	Transformer
R14	49.9	Resistor
R16	49.9	Resistor
R17	49.9	Resistor
R26	49.9	Resistor
C18	0.1uF	Capacitor
L2	BEAD	Inductor

BOM DONE

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Ethernet PHY

The schematic diagram illustrates the wiring for an Ethernet PHY (ENC28J60-1/ML) IC. The IC is connected to a +3.3V power supply and GND. Key connections include:

- Power and Ground:** VDD (24) and VDDPLL (16) are connected to +3.3V. VSS (26) and VSSPLL (17) are connected to GND.
- Reset:** RESET (6) is connected to +3.3V through a 1k resistor (R3).
- Crystal Oscillator:** A 25MHz crystal (Y1) is connected between OSC1 (19) and OSC2 (20). OSC1 is also connected to +3.3V (VDDOSC, 21). OSC2 is connected to GND (VSSOSC, 18). 16pF capacitors (C2, C3) are connected from the crystal nodes to GND.
- Control Signals:** SI (3), SO (2), SCK (4), CS (5), INT (28), and WOL (1) are connected to their respective external signals (ENC_SI, ENC_SO, ENC_SCK, ENC_CS, ENC_INT, ENC_WOL).
- LEDs:** LEDA (23) and LEDB (22) are connected to LEDA_CATH and LEDB_CATH.
- Transceiver Pins:** VDDRX (15), TPIN+ (9), TPIN- (8), VSSRX (7), VDDTX (11), TPOUT+ (13), TPOUT- (12), and VSSTX (14) are connected to +3.3V or +3.3V GND.
- Resistor and Capacitor Network:** RBIAS (10) is connected to +3.3V through a 2.32k resistor (R8). VCAP (25) is connected to +3.3V through a 10uF capacitor (C11). A series of four 0.1uF capacitors (C7, C8, C9, C10) are connected in parallel between +3.3V and GND.

Raspberry Pi Header

sheet56B34F8B

5V_SUPPLY 5V_MCU

File: ../../../../tmp/git/PoEPi-hardware/ws2812-controller/IdealDiode.sch

P1

3 GPIO2/I2C1_SDA VPP_(5v) 2

5 GPIO3/I2C1_SCL VPP_(5v) 4

8 GPIO14/UART_TXD VDD_(3.3v) 1 X

10 GPIO15/UART_RXD VDD_(3.3v) 17 X

ENC_SI 19 GPIO10/SPL_M0SI GND 6

ENC_SO 21 GPIO9/SPL_MISO GND 9

ENC_SCK 23 GPIO11/SPL_SCLK GND 14

ENC_CS 24 GPIO8/SPL_CE0 GND 20

26 GPIO7/SPL_CE1 GND 25

WS2812_DO 7 GPIO4/GPCLK0 GND 30

29 GPIO5 GND 34

31 GPIO6 GND 39

32 GPIO12

33 GPIO13

36 GPIO16

11 GPIO17 ID_SC 28

12 GPIO18 ID_SD 27

35 GPIO19

37 GPIO20

40 GPIO21

15 GPIO22

16 GPIO23

18 GPIO24

ENC_INT 22 GPIO25

37 GPIO26

13 GPIO27

Raspberry_Pi+_Conn

[illegible]

Linear Regulator

The schematic diagram illustrates a linear regulator circuit. The input voltage is +5V, connected to the VIN pin (pin 3) of the LM1117IMPX-ADJ (U3). The output voltage is +3.3V, connected to the VOUT pin (pin 2) of U3. The ADJ/GND pin (pin 1) is connected to ground through a 300 ohm resistor (R13). The output of the regulator is connected to a 240 ohm resistor (R15) and a 0.1uF capacitor (C19) in parallel, which is then connected to a 1uF capacitor (C20) in parallel with the output. The output is also connected to a 360 ohm resistor (R18) and an LED (D6) through a 360 ohm resistor (R18).

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Title: Raspberry Pi – Power Over Ethernet & PHY		
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B						
C						
D						
	1	2	3	4	5	6

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